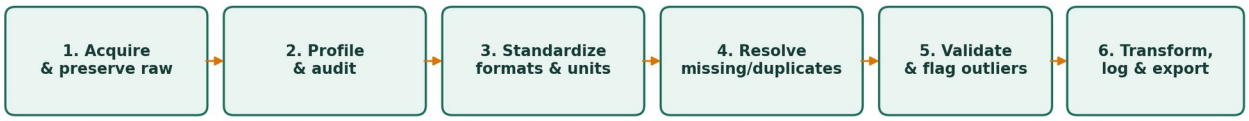
WEEK 2 | DATA SCIENCE PROJECT

DATA CLEANING & PREPROCESSING STRATEGY

Agribusiness dataset | Proposed strategy and reproducible example

PROPOSED AGRIBUSINESS PREPROCESSING PIPELINE



Each step writes a log; source values remain recoverable. Invalid records are flagged for review, not silently erased.

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Document Overview

Purpose

This document proposes a practical, auditable cleaning and preprocessing strategy for a hypothetical agribusiness dataset. It defines how common data-quality problems will be detected, handled, documented, and tested before analysis or forecasting.

Executive Summary

Agricultural datasets combine observations from farms, weather stations, market yards, surveys, and administrative records. Records may contain missing rainfall values, inconsistent district names, mixed measurement units, duplicate transactions, invalid dates, recording mistakes, or unusual values that could be genuine events. If these issues are not addressed systematically, summary statistics and models may be biased or misleading.

The proposed workflow preserves the original file, profiles fields, standardizes categories and units, resolves duplicates using a documented record grain, handles missing values according to variable meaning, flags outliers for review, checks domain logic, creates analytical features, validates outputs, and records every change. The example CSV and Python script are supplied as reproducible learning materials.

Item	Proposal
Dataset type	Hypothetical farm-level crop observation dataset
Example fields	Farm, date, district, crop, season, area, production, yield, rainfall, price, fertilizer intensity
Primary outcome fields	Production and yield; do not casually impute these labels for supervised prediction
Principle	Correct only when evidence supports a correction; otherwise flag, retain and document
Expected outputs	Cleaned table, issue log, data dictionary, validation summary and reproducible script

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1. Introduction and Objectives

Data cleaning and preprocessing transform collected data into a consistent, usable, and traceable analytical dataset. In agribusiness, a small recording problem can change apparent crop productivity, input use, market-price comparisons, or climate relationships. For example, area entered in acres but interpreted as hectares can distort calculated yield; a date parsed in the wrong order can assign observations to the wrong season; a valid extreme rainfall event can be mistaken for an error if outliers are deleted automatically.

1.1 Objectives

- Identify likely quality issues in farm, crop, weather, and market observations.
- Specify clear rules for missing values, duplicates, formats, units, outliers, and logical inconsistencies.
- Create a step-by-step process that separates source correction from analytical transformation.
- Prevent target leakage and inappropriate imputation in future forecasting models.
- Preserve reproducibility through raw-data retention, versioned code, parameters, issue logs, and validation checks.
- Demonstrate the strategy using a labelled hypothetical dataset and a runnable Python script.

1.2 Scope and assumptions

This is a proposed strategy, not a claim that the synthetic example represents real farms or actual prices. The sample observations were deliberately constructed to show common problems. Real implementation must confirm field definitions with the data owner, use crop- and region-specific validation limits, and follow source terms and privacy requirements. Where a value cannot be verified, it is flagged rather than silently modified.

Important distinction

A data-quality flag is not the same as proof that a value is wrong. A high yield, high price, or rainfall extreme can be real. The process should investigate context, units, and provenance before excluding an observation.

2. Hypothetical Dataset and Data-Quality Risks

The example dataset represents one farm-crop-season observation per farm and date. The proposed record grain must be agreed before duplicate checks: in this demonstration, Farm_ID + parsed Observation_Date + Crop + Season is the candidate business key. If a real table records multiple plots, harvests, or market transactions per day, the key must be expanded.

Field	Type / unit	Validation rule
Record_ID	Text identifier	Must be unique; never use as a predictive feature by default
Farm_ID	Categorical ID	Non-empty; validate against approved farm registry if available
Observation_Date	Date	Parse with source-specific format; invalid dates flagged
District, Crop, Season	Categorical text	Trim whitespace and map approved spelling variants

Field	Type / unit	Validation rule
Area_Value + Area_Unit	Numeric + source unit	Convert to hectares; area must be positive for yield calculation
Production_Value + Production_Unit	Numeric + source unit	Convert to tonnes; negative production is invalid
Reported_Yield_t_ha	Numeric, t/ha	Compare with production ÷ area where both are valid
Rainfall_mm	Numeric, mm	Non-negative; compare with station and time period
Farmgate_Price_Value + Price_Unit	Numeric + source unit	Convert to INR/quintal only after unit is confirmed
Fertilizer_kg_ha	Numeric, kg/ha	Non-negative; clarify whether value is product mass or nutrient mass

2.1 Common quality issues

Issue	Example	Risk if ignored
Missing value	Rainfall blank for a farm-date	Biased means, reduced rows, or hidden seasonal gaps
Format inconsistency	Davangere / Davanagere / extra spaces	Duplicate categories and fragmented totals
Unit inconsistency	1 acre mixed with hectares; kg mixed with tonnes	Incorrect area, production, price, and yield comparisons
Duplicate record	Same farm, date, crop and season entered twice	Double-counted area or production
Outlier / anomaly	Reported grain yield of 50 t/ha in example	Distorted averages or model fit if unit/error not investigated
Domain violation	Negative rainfall or fertilizer quantity	Physically impossible or malformed input
Logical contradiction	Area = 0 while production > 0	Yield division invalid; fields need source verification
Invalid date	31-14-2025	Incorrect time-series placement or lost temporal ordering
Category missing	No crop recorded	Unsafe crop-level aggregation or fabricated category assignment

3. Step-by-Step Data Cleaning Plan

Step	Action and output
1. Acquire and preserve	Save the original downloaded file as read-only. Record source URL/owner, extraction date, filters, license, file size and a checksum if possible. Never overwrite the raw file.
2. Profile the dataset	Record row/column counts, data types, unique categories, missingness by field, duplicate candidates, ranges, percentiles, and distributions. Generate a baseline data-quality report before edits.
3. Confirm schema and record grain	Define each field, unit, acceptable domain, primary key and observation grain with a data dictionary. Decide which rows are observations, summaries, or transactions.
4. Standardize text and identifiers	Trim whitespace; standardize casing; map only verified synonyms to canonical labels; preserve raw text in audit columns. Do not merge ambiguous place names without a reference list.
5. Parse dates and units	Use source-aware date formats and explicit unit conversions. Unparsed dates and unknown units become flagged missing standardized fields, not guessed values.
6. Detect duplicates	Check exact duplicates and candidate business-key duplicates after date/category normalization. Compare conflicting duplicates with the source; keep a traceable decision log.
7. Assess missing values	Measure amount and pattern of missingness. Distinguish not collected, not applicable, suppressed, invalid parse, and true zero. Search for source correction before imputation.
8. Apply variable-specific treatments	Use defensible imputations for eligible predictors; leave target labels missing for supervised training or exclude those rows from that target-specific task. Add imputation indicators.
9. Validate domain and cross-field logic	Check non-negative area/production/rainfall/price; positive area for yield; production/area agreement; crop-season compatibility; and plausible ranges informed by metadata.
10. Investigate outliers	Use visual inspection, IQR or robust z-scores where sample size supports them, seasonal/crop peer comparisons, and source checks. Flag before removing or winsorizing.
11. Build analysis features	Calculate yield only from valid area and production, aggregate to agreed time/place level, encode categories, and scale numeric features only where needed by the model.
12. Validate and export	Run acceptance checks, compare before/after counts and summary distributions, write cleaned data and issue log, and document every exception in the run summary.

4. Rules for Specific Data-Quality Problems

4.1 Missing-value strategy

Variable class	Preferred action	Avoid
Area, production or yield target	Ask source to correct. If unresolved, retain as missing and exclude only from analyses requiring that target; report exclusions.	Do not fill a missing target with the overall mean or a guessed value.
Rainfall/weather predictor	Prefer a validated weather station/grid for matching date and location. If unavailable, consider within-series interpolation or crop-season median only if justified; add imputation flag.	Do not interpolate across large gaps, seasons, stations, or extremes without rationale.
Numeric input predictor	Use median or stratified median (crop/region/season) if missingness and sample size support it; fit imputation on training data only.	Do not compute imputation values using future/test data.
Categorical predictor	Use an explicit “Unknown/Not recorded” category where meaningful, or mode only when justified.	Do not use mode to guess a missing crop label for crop-specific conclusions.
Structural missing / not applicable	Keep as a separate status or retain null with a reason code.	Do not treat “not applicable” as zero.

Decision rule

Before imputation, answer: Why is it missing? Is there a trusted source to retrieve it? Is it a model input or the outcome to predict? How much is missing and in what pattern? What evidence supports the chosen method? Record the answers in the cleaning log.

4.2 Inconsistent formats and units

Raw example	Canonical representation	Rule
" rice ", "Rice", "RICE"	Rice	Trim and title/canonical case; use approved crop dictionary.
Davanagere / Davangere	One verified district label	Map only if reference source confirms they refer to the same entity.
1 acre	0.404686 ha	Area_ha = Area_acre × 0.404686. Retain source unit and source value.
1,200 kg	1.2 tonnes	Production_t = Production_kg ÷ 1,000.
₹22 per kg	₹2,200 per quintal	Price_INR_quintal = Price_INR_kg × 100; verify price definition first.

Raw example	Canonical representation	Rule
12/06/2025	2025-06-12 if source defines D/M/Y	Use explicit source date format; ambiguous dates must be resolved, not guessed.

4.3 Duplicates

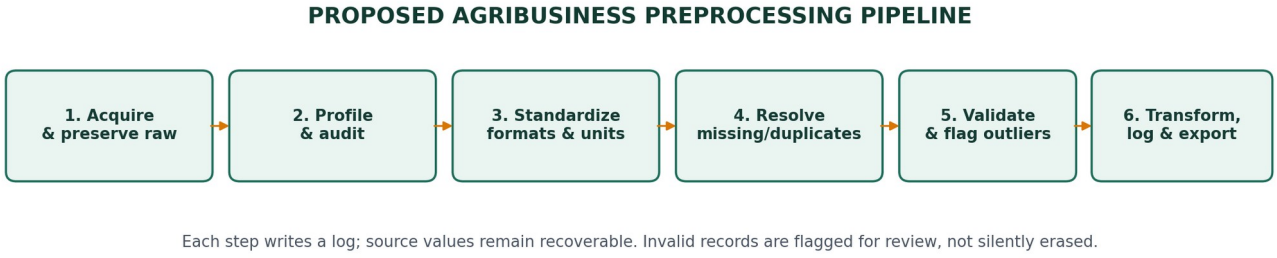
Exact duplicate rows can often be removed from an analytical copy after confirmation. Business duplicates require more care: two rows may share farm/date/crop but refer to separate plots or harvests. The key must match the real record grain. Keep dropped row IDs and the reason, and retain the raw source. If duplicate values conflict, reconcile against the origin rather than choosing whichever is convenient.

4.4 Outliers and anomalies

- Start with histograms, boxplots, time-series plots, percentiles, and crop-region-season comparisons.
- Use IQR fences ($Q1 - 1.5 \times IQR$, $Q3 + 1.5 \times IQR$) or robust z-scores as screening tools, not automatic proof of an error.
- Validate units, decimal placement, sensor calibration, geography, weather events, crop variety, season, and source documents.
- If verified valid, retain the value and consider robust statistics or suitable transformations. If a source error is proven, correct with provenance; if unresolved, flag and run sensitivity analysis.
- Winsorization or exclusion must be justified, parameterized, and reported. Do not remove observations solely because they worsen a model.

5. Proposed Preprocessing Framework

The workflow below provides the recommended sequence. The source data remains immutable; processed datasets and analysis-ready subsets are new versioned outputs. Each stage should record its row count, columns changed, flags raised, and reason for each transformation.



5.1 Feature engineering and calculations

Feature	Formula / method	Guard condition
Area_ha	Area_acre × 0.404686; hectares unchanged	Known source unit; area not missing
Production_t	Production_kg ÷ 1,000; tonnes unchanged	Known source unit; production non-negative
Calculated_Yield_t_ha	Production_t ÷ Area_ha	Area_ha > 0 and both inputs valid
Price_INR_quintal	Price_INR_kg × 100	Known price unit and same price definition/grade
Season/year features	Extract from validated observation date and declared season calendar	Date parsed and time-zone/calendar assumptions clear
Imputation indicator	1 if field was imputed; 0 otherwise	Create before imputation so provenance remains visible

5.2 Encoding and scaling

- One-hot encoding can represent crop, district, season, or soil category for many statistical and machine-learning models. Avoid arbitrary numeric codes for nominal categories unless the model explicitly treats them as categories.
- Standardization uses z = (x − training mean) / training standard deviation; min-max scaling maps values to a chosen interval, often [0,1]. Learn parameters on training data only and apply them to validation/test data.
- Scaling is important for distance-based and many gradient/regularized models, but tree-based models generally do not require numeric scaling. Choose preprocessing based on the intended model, not as an automatic step for every dataset.
- For forecasting, split by time rather than randomly mixing dates. Do not use future weather, future price, or outcome-derived features unavailable at prediction time.

5.3 Filtering and analytical subsets

Maintain a full cleaned-reference table with quality flags and create purpose-specific subsets. For example, yield analysis may require a valid crop, valid date, positive area, valid production, and a resolved or acceptable yield check. Market-price analysis may require a declared commodity grade, market, date, and unit. Every filter should produce a row count and exclusion-reason summary.

6. Worked Hypothetical Examples

Synthetic data warning

All values in this section and in hypothetical_agri_raw.csv were invented for demonstration. They are not observations of real farms and must not be presented as official agricultural statistics.

Raw record	Issue detected	Proposed treatment
R002: 1 acre; 1,200 kg maize	Area/production units differ from standard	1 acre → 0.404686 ha; 1,200 kg → 1.2 t; calculated yield ≈ 2.97 t/ha if the values are verified.
R001 and R004: same farm/date/crop/season	Potential duplicate business event	Flag using candidate business key; keep one in analytical copy only after confirming it is a duplicate; log removed ID.
R007: rainfall missing	Missing predictor	Use validated station data if available; in the demo script, use a crop-season median when available and mark it imputed.
R005: 2 ha; 100 t production; 50 t/ha	Extremely unusual grain yield in demo	Flag for review. Check decimal/unit/source. Do not auto-delete; crop-specific plausibility limits must be established from trusted references.
R008: 0 ha; 3 t production	Logical contradiction	Do not calculate yield. Verify area and production at the source; exclude from yield modeling until resolved.
R009: date 31-14-2025	Unparseable date	Keep raw date, record parse failure and exclude from time-series analysis until corrected.
R010: rainfall −15; fertilizer "high"	Domain violation and non-numeric input	Preserve raw record; mark rainfall missing in analytical field and request correction of fertilizer value.
R014: ₹24/kg	Price unit differs	Standardize to ₹2,400/quintal if the quoted price basis is otherwise comparable.

6.1 Mini calculation

For R002 (after unit conversion), area = 1 × 0.404686 = 0.404686 ha. Production = 1,200 ÷ 1,000 = 1.2 tonnes. If both inputs are verified and area is positive, calculated yield = 1.2 ÷ 0.404686 ≈ 2.97 tonnes/ha. The reported yield is missing, so the calculated value should be labelled as derived—not copied into the reported source field.

7. Reproducibility and Data Governance

- Keep immutable raw data and write all transformations to new files; use versioned folders such as raw/, interim/, processed/, outputs/ and logs/.
- Store the script/notebook, package versions, parameters, source URL, retrieval date, input filename, and expected output schema alongside the data.

- Make mappings, thresholds, unit conversions, imputation methods, and duplicate keys explicit configuration rather than undocumented edits.
- Write a quality issue log with Record_ID, field, issue, action, and status; retain row-level flags in the cleaned data.
- Re-run from raw input in a clean environment and compare output row counts, missingness, unit conversions, and key summary values.
- Use deterministic methods where possible. If a randomized method is used, set and document a random seed.
- Do not include private farmer names, phone numbers, or precise farm coordinates in public examples unless permitted and necessary; minimize and protect personal data.

7.1 Suggested quality acceptance checks

Check	Acceptance rule for this strategy	Evidence to save
Schema	All expected columns present; standardized fields have documented types and units	Schema/data dictionary report
Dates	No unparseable date enters time-series/model subset without documented exception	Invalid-date issue list
Units	Every analytical numeric unit is canonical or explicitly flagged	Conversion log and unit counts
Duplicates	Duplicate business keys reviewed; final duplicate count explained	Duplicate-resolution log
Missingness	Missing count and percentage reported per key field before/after imputation	Missingness summary
Domain rules	Negative values, zero denominators and logical contradictions flagged	Validation report
Derived yield	Equals production_t / area_ha within tolerance for valid rows	Automated assertion results
Provenance	Each transformation traceable to script and configuration	Versioned code and run log

8. Execution Timeline and Evaluation Rubric

This seven-day plan is a proposed schedule for the Week 2 task. Actual dates may be assigned according to the course submission calendar.

Day	Task / milestone	Completion evidence
Day 1	Confirm schema, observation grain, source and data dictionary	Approved field definitions and units
Day 2	Profile raw dataset and calculate quality baseline	Profiling report: missingness, ranges, duplicates
Day 3	Standardize formats, categories, dates and units	Scripted mappings and conversion tests
Day 4	Define missing-value and duplicate treatments	Decision log with variable-specific rules
Day 5	Run domain checks, cross-field logic and outlier review	Quality issue log and flagged-record review
Day 6	Build preprocessing pipeline and reproducible example	Script/notebook, cleaned sample CSV, validation checks
Day 7	Review, document limitations and package deliverables	Final DOCX, data dictionary, workflow and reproducibility notes

8.1 Evaluation rubric (suggested, not official)

Criterion	Weight	Excellent performance (4–5/5)
Coverage of quality issues	15%	Explains missingness, formats/units, duplicates, outliers, logical/domain issues and their risks.
Specific treatment rules	25%	Gives variable-aware, justified steps and distinguishes corrections, imputations, flags and exclusions.
Workflow and logical order	15%	Clear sequence from raw-data preservation through validation and export.
Hypothetical examples and calculations	10%	Correct conversions and before/after examples, clearly marked synthetic.
Reproducibility	15%	Raw data, script, environment, parameters, logs, and outputs can be rerun and audited.
Data validation / acceptance checks	10%	Measurable tests and evidence are provided.
Clarity, structure and references	10%	Professional organization, readable tables, limitations and credible documentation links.
TOTAL	100%	A complete, testable strategy rather than generic descriptions.

9. Conclusion

A reliable preprocessing strategy is not merely a list of cleaning functions. It is a documented decision process that respects the meaning and context of agricultural fields. This report recommends retaining raw data, confirming schema and units, measuring quality before and after processing, using cautious variable-specific treatments for missing values, reviewing rather than automatically deleting outliers, validating cross-field logic, preventing data leakage, and preserving code and issue logs. The hypothetical example demonstrates how these decisions can be made consistently and reproduced. Before using the strategy on production data, the data owner should confirm the observation grain, crop-specific ranges, source definitions, privacy restrictions, and downstream analysis objective.

References and Technical Resources

1. **pandas documentation — Working with missing data** https://pandas.pydata.org/docs/user_guide/missing_data.html
2. **scikit-learn documentation — Imputation of missing values** https://scikit-learn.org/stable/modules/impute.html
3. **scikit-learn documentation — Preprocessing data** https://scikit-learn.org/stable/modules/preprocessing.html
4. **NIST/SEMATECH e-Handbook — Identifying outliers** https://www.itl.nist.gov/div898/handbook/eda/section3/eda35h.htm
5. **FAOSTAT** https://www.fao.org/faostat/en/#data
6. **World Bank — Agriculture and Rural Development** https://data.worldbank.org/topic/agriculture-and-rural-development
7. **DA&FW / Directorate of Economics and Statistics** https://desagri.gov.in/

Appendix A. Reproducible Example

The companion ZIP includes hypothetical_agri_raw.csv and clean_agri_data.py. Run them in the same folder after installing pandas. The script generates cleaned_agri_example.csv and quality_issues_log.csv. The sample includes intentional inconsistencies for teaching purposes.

Run instructions

1. Install Python 3.10+ and pandas. 2. Extract all files into one folder. 3. Run: python clean_agri_data.py 4. Inspect cleaned_agri_example.csv and quality_issues_log.csv. The script preserves its input and records flagged issues. Output values are synthetic demonstration data.

Core validation logic (illustrative)

```
# Calculate yield only when units are standardized and area is positive
valid = area_ha.notna() & (area_ha > 0) & production_t.notna() & (production_t >= 0)
calculated_yield_t_ha = production_t[valid] / area_ha[valid]

# Flag a reported/calculated mismatch; investigate rather than auto-delete
relative_gap = abs(reported_yield - calculated_yield_t_ha) / calculated_yield_t_ha
flag_for_review = relative_gap > 0.10  # example review threshold; justify for each use case

# Fit imputers/scalers only on training data, then transform validation/test data
```

Final reminder

The 10% discrepancy and >20 t/ha example flag in the companion script are demonstrative starting points, not universal agronomic thresholds. Calibrate all thresholds using crop, season, district, source quality, and reliable reference statistics.